

Diagram Gallery

Companion to “Tool Support for Causal Reasoning via String Diagram Surgery”

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This companion document collects the full-resolution diagrams produced by the tool for the running *commute* example, omitted from the page-limited submission for space. Each figure corresponds to a stage of the pipeline described in the paper (Section 3); diagrams from Step 2 onward are direct **Chyp exports** written to the `trace/` directory by the tool, so they show exactly what a user inspects when running the pipeline. (Chyp uses a limited character set, so some symbols are mangled in the labels.) The raw `.chyp` sources are under `trace/commute/` in the repository.

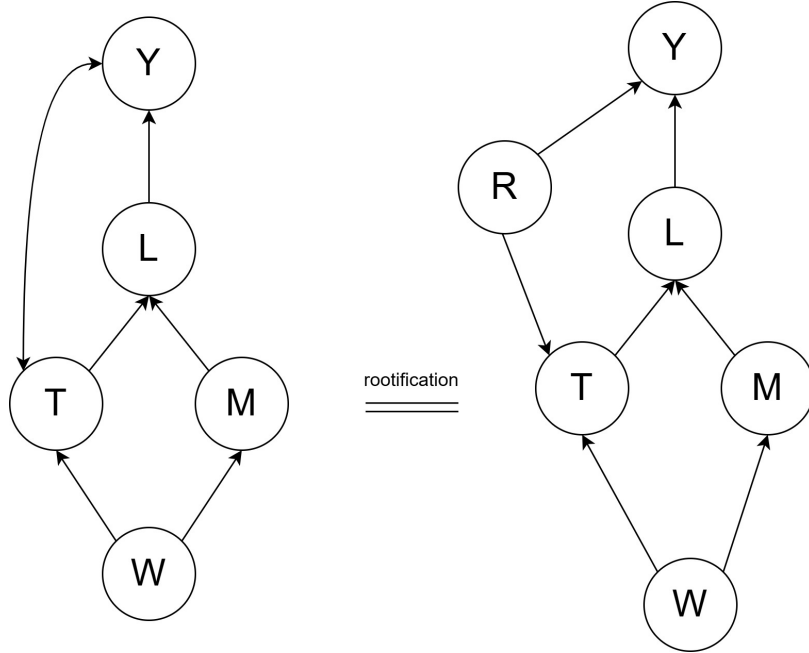


Figure 1: **Stage 1 — ADMG compilation (rootification)**. The input Acyclic Directed Mixed Graph for the commute model (left) and its rootified form (right), in which the bidirected edge $T \leftrightarrow Y$ is replaced by a fresh latent root R feeding both T and Y .

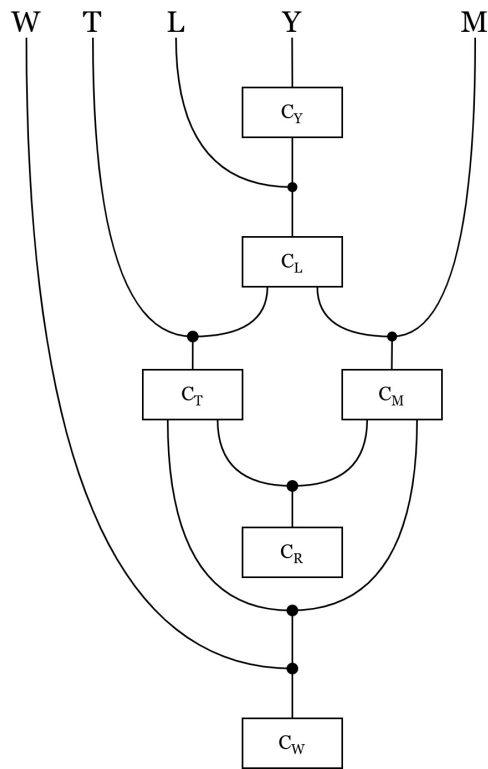


Figure 2: **Stage 1 (cont.) — compiled string diagram.** The commute model as a Catlab wiring diagram: each variable becomes a mechanism box, copy dots duplicate values feeding several mechanisms, and the shared latent root realises the confounding $T \leftrightarrow Y$.

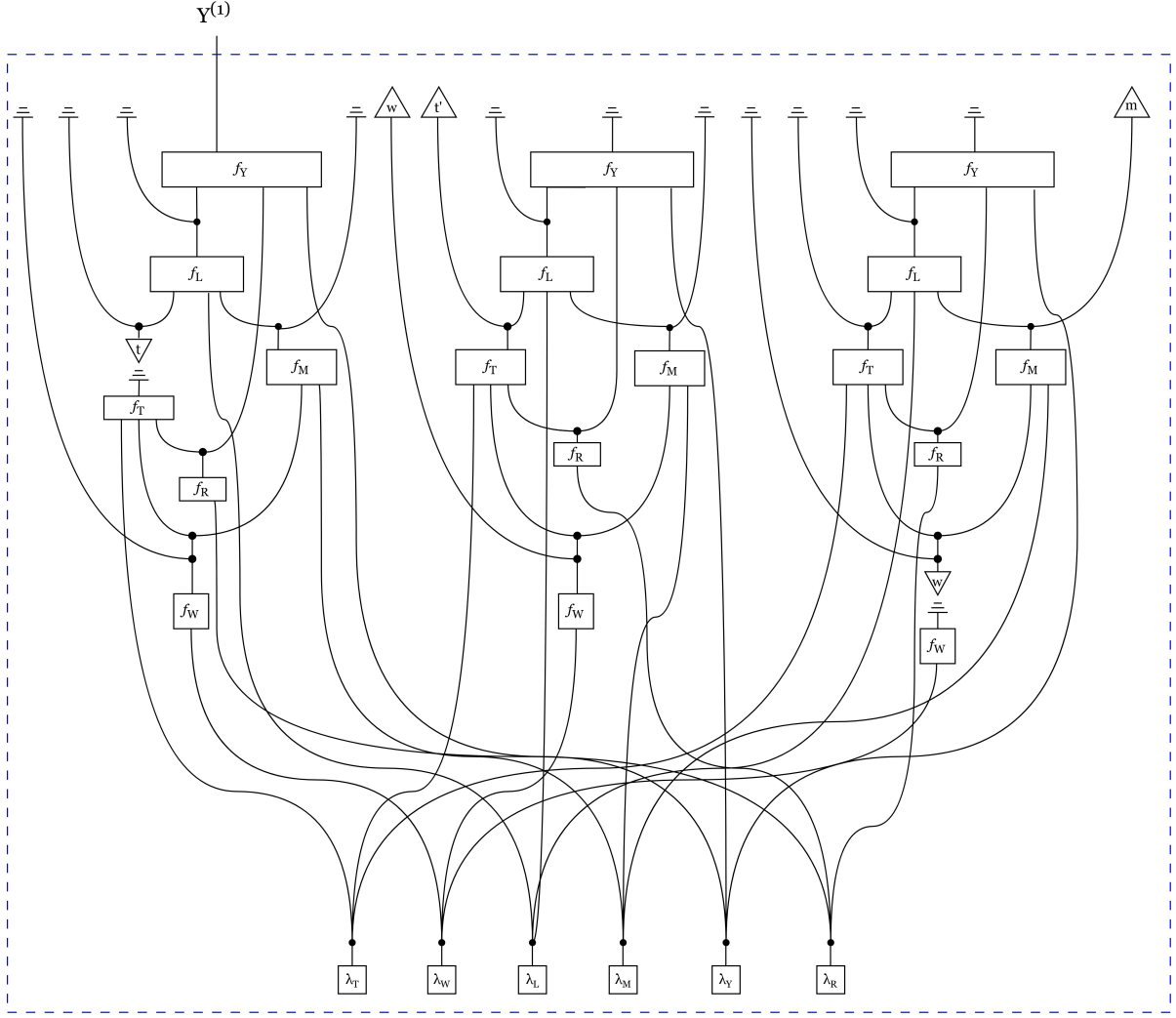


Figure 3: **Stage 2 — parallel-world (twin) construction.** The base diagram lifted into a multiverse: each mechanism is duplicated per world while the exogenous sources are shared, so the factual and counterfactual worlds stay coupled through the common latent structure. Surgery $\text{do}(T=t)$ is applied only in the counterfactual world.

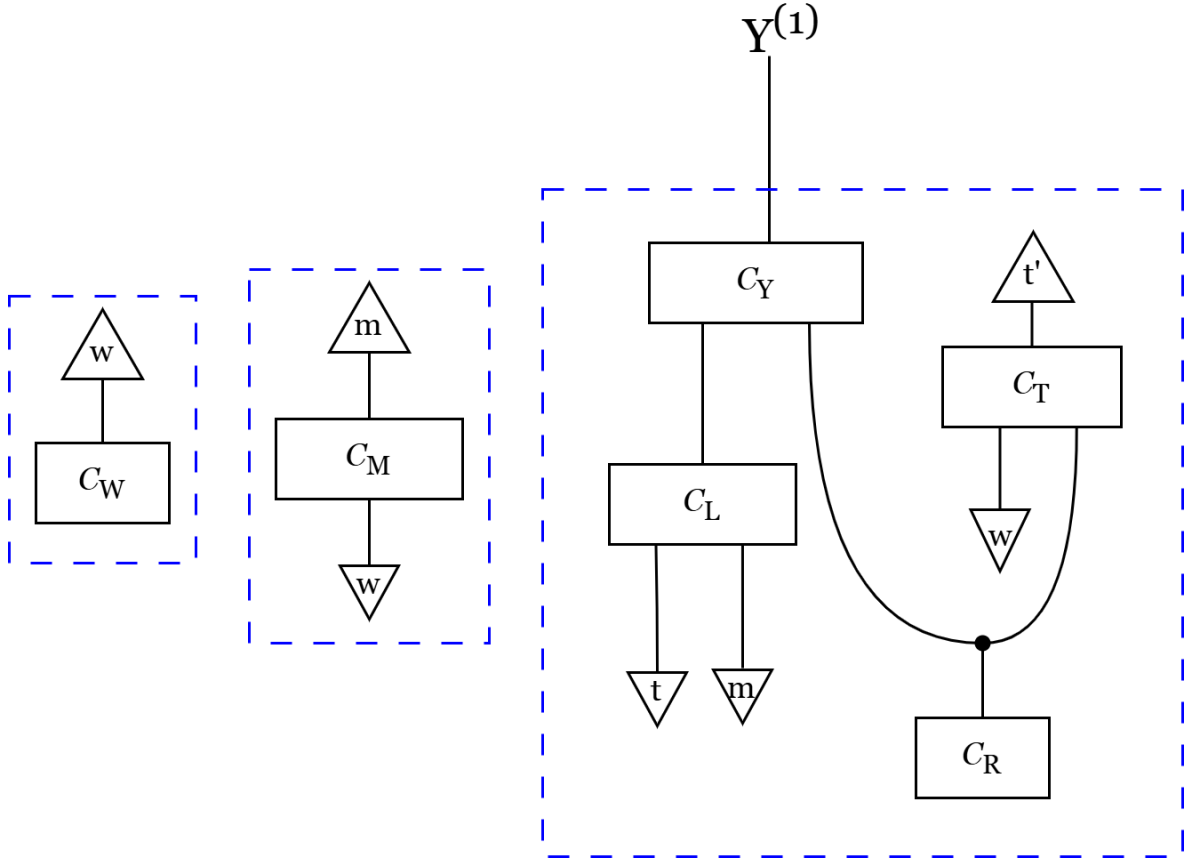


Figure 4: **Stage 3 — simplified diagram.** The diagram after the four rewrite passes (pruning, observation separation, deterministic-mechanism merging, exogenous-source absorption) reduce it to canonical form.

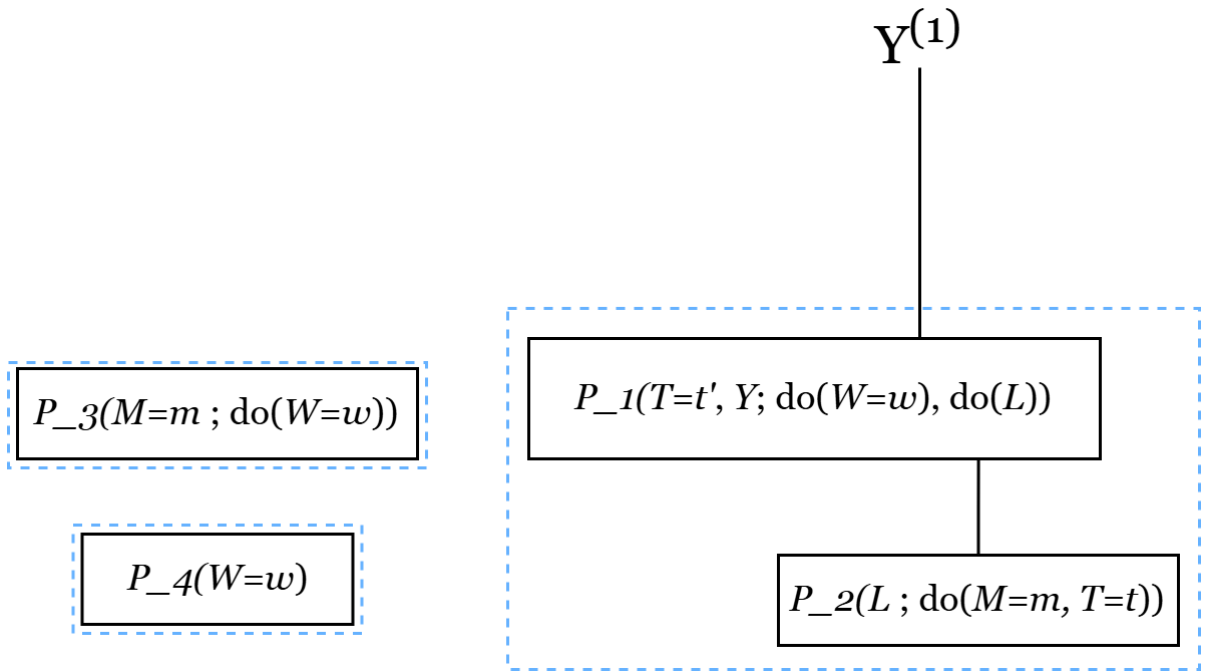


Figure 5: **Stages 4–5 — identified estimand.** The R-fragments rewritten into interventional probability boxes and rendered as the final estimand. A residual unabsorbed exogenous source would instead signal a non-identifiable query.